

3.3 Halogen Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Hard

Time allowed: 20

Score: /10

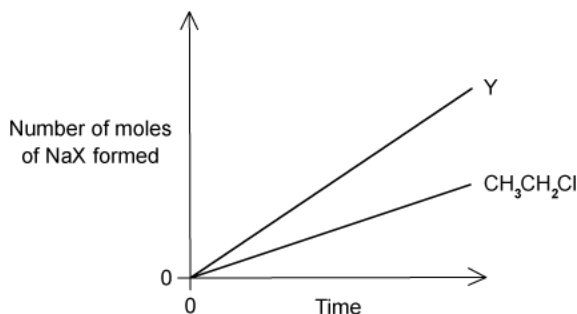
Percentage: /100

Question 1

When halogenoalkanes, RX , are hydrolysed with $NaOH$, the corresponding sodium halide, NaX , is produced.



A student investigated the amount of NaX produced by hydrolysing CH_3CH_2Cl and another halogenoalkane, Y . In a given time the amount of sodium halide formed was greater with Y than with CH_3CH_2Cl .



Which compound(s) could be Y ?

- 1 $ClCH_2CH_2Cl$
- 2 CH_3CH_2Br
- 3 CH_3CH_2I

- A. 1, 2 and 3
 B. 1 and 2
 C. 2 and 3
 D. 1 only

[1 mark]

Question 2

When an isomer Y of molecular formula C_4H_9Br undergoes hydrolysis in aqueous alkali to form an alcohol C_4H_9OH , the rate of reaction is found to be unaffected by changes in the concentration of OH^- ions present.

Which is the most likely molecular structure of Y ?

- A. $CH_3CH_2CH_2CH_2Br$
 B. $CH_3CH_2CHBrCH_3$
 C. $(CH_3)_2CHCH_2Br$
 D. $(CH_3)_3CBr$

[1 mark]

Question 3Which statements are true for an S_N2 reaction?

- 1 One bond is broken as another bond is formed.
- 2 The formation of a transition state involves the collision of two molecules or ions.
- 3 A carbon atom in the transition state is bonded, either fully or partially, to five other atoms.

A. 1, 2 and 3

B. 1 and 2

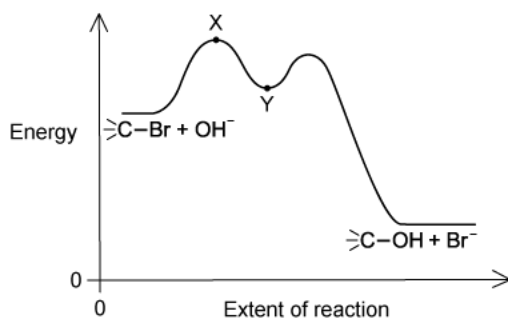
C. 2 and 3

D. 1 only

[1 mark]

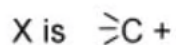
Question 4

A tertiary bromoalkene, indicated here by >C-Br , reacts with aqueous NaOH . The mechanism has the reaction pathway below.

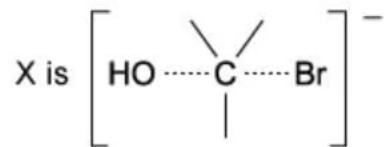


Which point in the diagram is correctly identified?

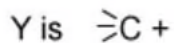
A.



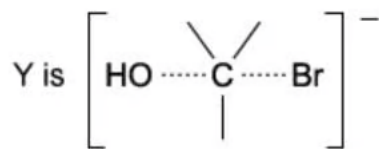
B.



C.



D.



[1 mark]

Question 5

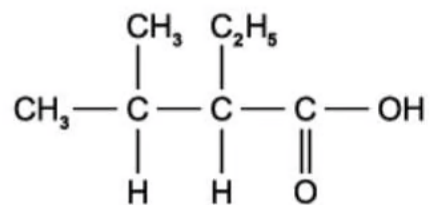
When a solution of potassium cyanide in ethanol is heated under reflux with 1-chloropropane, what is the organic product of the reaction called?

- A. Butanenitrile
- B. 1-chloropropanenitrile
- C. Ethanenitrile
- D. Propanenitrile

[1 mark]

Question 6

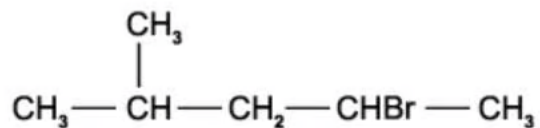
The compound 2-ethyl-3-methylbutanoic acid is used as a flavouring in some food.



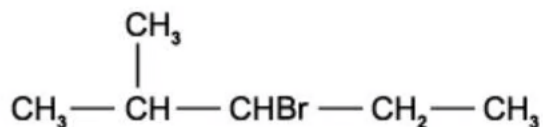
2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid when heated under reflux with sodium cyanide in ethanol, followed by acid hydrolysis of the reaction product?

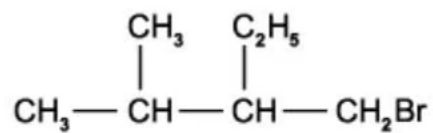
A.



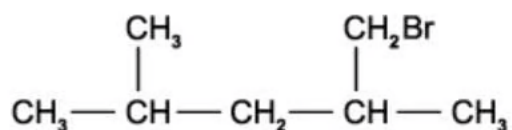
B.



C.



D.



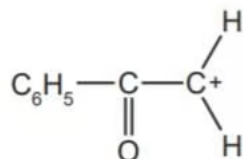
[1 mark]

Question 7

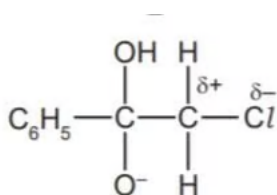
When phenacyl chloride, $\text{C}_6\text{H}_5\text{COCH}_2\text{Cl}$, is reacted with aqueous NaOH , the substitution reaction follows an $\text{S}_{\text{N}}2$ mechanism.

Which structure represents a species formed during the reaction?

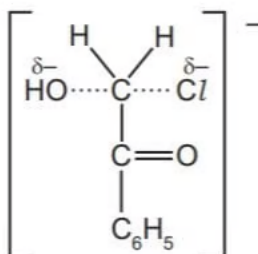
A.



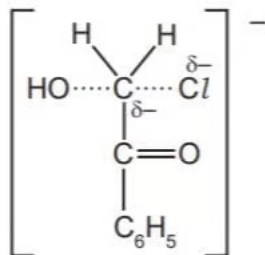
B.



C.



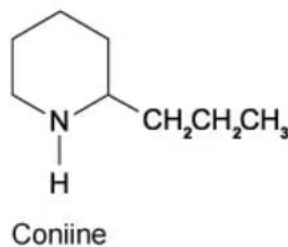
D.



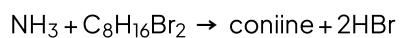
[1 mark]

Question 8

Coniine is the major constituent of the poison 'oil of hemlock'.



Coniine can be synthesised by reacting ammonia with a dibromo compound, **X**.



X

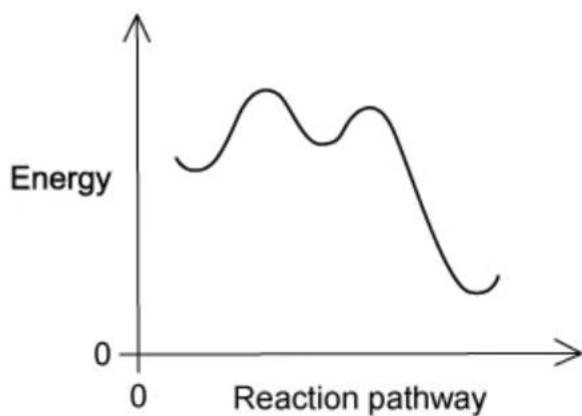
What is the name of compound **X**?

- A. 1,1-dibromo-2-propylcyclopentane
- B. 1,2-dibromo-2-propylcyclopentane
- C. 1,4-dibromooctane
- D. 1,5-dibromooctane

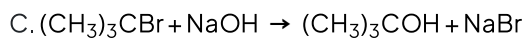
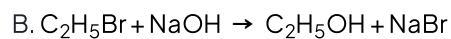
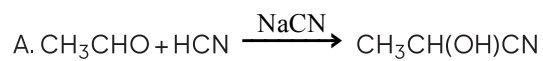
[1 mark]

Question 9

A reaction pathway diagram is shown.



Which reaction does **not** have such a profile?



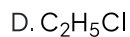
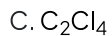
D.



[1 mark]

Question 10

Which of the following molecules will have **no** permanent dipole?



[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- When comparing the rate of hydrolysis of haloalkanes, the trend, from most reactive to least reactive is:
 - Iodoalkanes > Bromoalkanes > Chloroalkanes
- This is because the rate of reaction increases as bond strength decreases.
 - In other words, the C-I bond is the weakest, so, it is broken most easily.
- $\text{ClCH}_2\text{CH}_2\text{C}/$ will produce two moles of sodium halide per one mole of compound Y whereas $\text{CH}_3\text{CH}_2\text{C}/$ will only produce one mole of sodium halide.
- Therefore, more bromoalkanes and iodoalkanes are likely to react in the given time and produce more sodium halide than with $\text{CH}_3\text{CH}_2\text{C}/$ and $\text{C}/\text{CH}_2\text{CH}_2\text{C}/$ will form more moles of sodium halide per mol of reactant.
 - Hence compound Y could be $\text{CH}_3\text{CH}_2\text{Br}$ or $\text{CH}_3\text{CH}_2\text{I}$ or $\text{C}/\text{CH}_2\text{CH}_2\text{C}/$

2

The correct answer is **D** because:

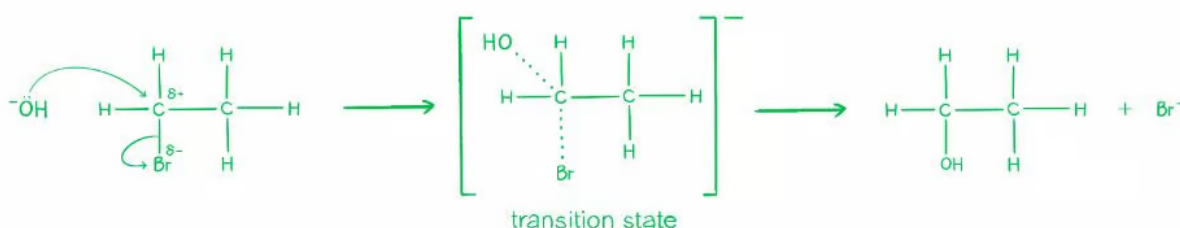
- The question tells us that when an isomer **Y** of molecular formula $\text{C}_4\text{H}_9\text{Br}$ undergoes hydrolysis in aqueous alkali, the rate of reaction is found to be unaffected by changes in the concentration of OH^- present.
 - This explains that the rate of reaction is unaffected by the presence of the nucleophile on the catalyst.
- Therefore, it is the example of an $\text{S}_{\text{N}}1$ reaction (**S**-ubstitution **n**-ucleophilic uni-molecular) in which the reaction depends only on the structure of the reactant not on the catalyst or the nucleophile.
- The Br is bonded to a tertiary carbon.
 - A carbon bonded to three other carbon atoms.

- This makes the bond more stable compared to if the Br was bonded to a primary or secondary carbon, so, it is unlikely for spontaneous substitution of Br for OH to occur.
- Hydrolysis occurs through reaction intermediates that make the making and breaking of C–OH and C–Br bonds respectively more favourable.
 - Therefore, out of the given options, $(\text{CH}_3)_3\text{CBr}$ is the correct answer.

3

The correct answer is **A** because:

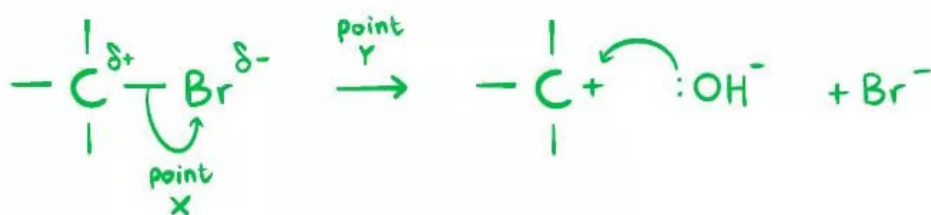
- In an $\text{S}_{\text{N}}2$ reaction (**S**-ubstitution **n**-ucleophilic bi-molecular), a bond is broken and a new bond is formed simultaneously in one intermediate step.
 - Therefore, statement 1 is correct.
- The nucleophile attacks the substrate on the opposite side to the leaving group.
- The formation and breaking of bond take place simultaneously (unlike the $\text{S}_{\text{N}}1$ reaction) and the rate of reaction depends on the concentration of both the nucleophile and the substrate.
- So, the formation of a transition state involves the collision of two molecules or ions.
 - Therefore, statement 2 is correct.
- The transition stage is represented by the square brackets – we can see the central carbon atom is bonded, either fully or partially, to five other atoms.
 - Therefore, statement 3 is correct.



4

The correct answer is **C** because:

- A tertiary halogenoalkane reacts with a hydroxide ion by a two-step mechanism.
- This mechanism is known as an S_N1 mechanism.
 - The '1' tells us that the rate of the reaction only depends on one reagent, in this case the concentration of the halogenoalkane, as shown in the first (slow) step of the mechanism.
- The first step in the mechanism is the breaking of the carbon-halogen bond - this forms a tertiary carbocation, as shown below:



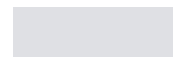
- Then, the tertiary carbocation is attacked immediately by the hydroxide ion.
- Reaction pathway diagrams display information about the energy changes during chemical reactions.
- The first bump corresponds to the slow step, as more energy is required here, and the second bump corresponds to the fast step, as less energy is required at this stage.
- Point X corresponds to the breaking of the C-Br bond.
- So, point Y corresponds to the intermediate step when the tertiary carbocation is formed.
 - Therefore, option **C** is correct.

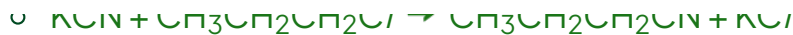
5

The correct answer is **A** because:

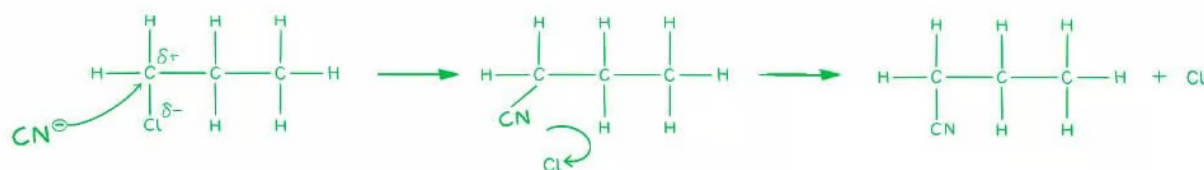
- The cyanide ion CN^- is a nucleophile and reacts with haloalkanes to produce a nitrile.

- This is a useful reaction for increasing the chain length by one carbon atom.
- Potassium cyanide + 1-chloropropane → butanenitrile + potassium chloride
 - $\text{KCN} + \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{KCl}$
- Therefore, **A** is correct.
- Mechanism:
 - **Step 1:** The cyanide ion attacks at the partially positive carbon of the dipole, leading to a high energy transition state.





- Therefore, **A** is correct.
- Mechanism:
 - **Step 1:** The cyanide ion attacks at the partially positive carbon of the dipole, leading to a high energy transition state.
 - **Step 2:** The chlorine atom leaves with its bonding electrons as a chloride ion.
 - **Step 3:** The final product is butanenitrile.



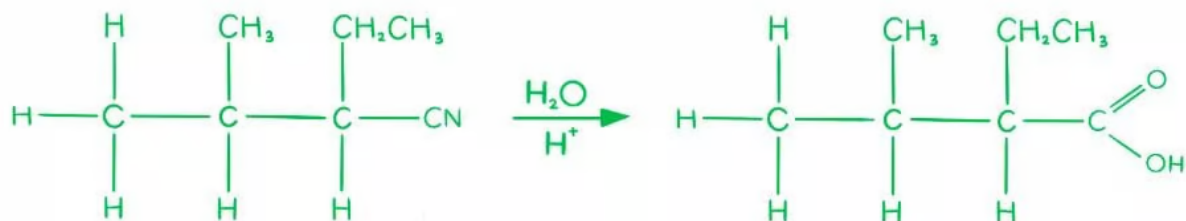
6

The correct answer is **B** because:

- When halogenoalkanes are heated under reflux with ethanolic sodium cyanide, there is an addition of a C atom across the chain when the halogen atom is substituted by CN.
- 2-ethyl-3-methylbutanoic acid has a total of 7 carbon atoms.
 - Hence the halogenoalkane used to synthesise should have 6 carbon atoms.
- Consider compound **B**:
- When heated under reflux with sodium cyanide in ethanol, the Br atom is substituted for CN:



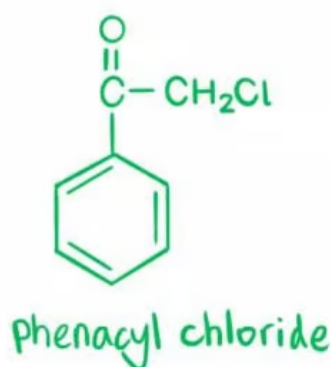
- The reaction of the CN group under acid hydrolysis will then produce a carboxylic acid.
- If we switch the CN and CH₂CH₃, or C₂H₅, groups, this is easier to visualise.



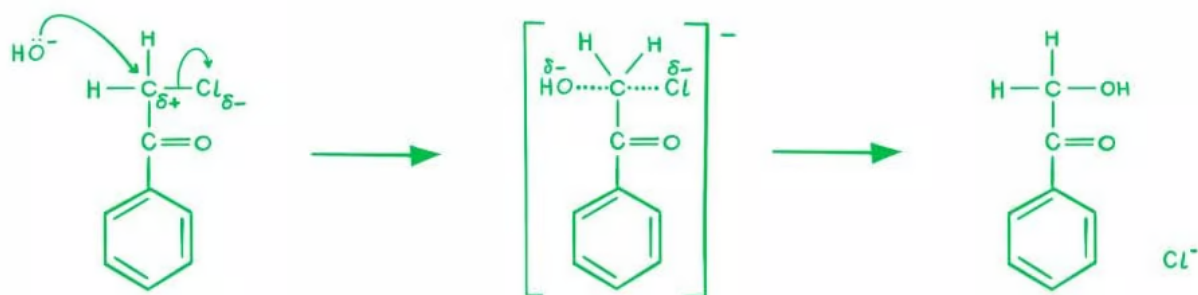
7

The correct answer is **C** because:

- The structure of phenacyl chloride is:



- When reacted with aqueous NaOH, the reaction proceeds as follows:

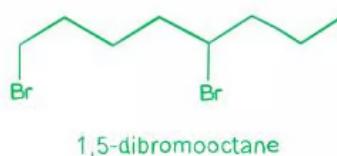
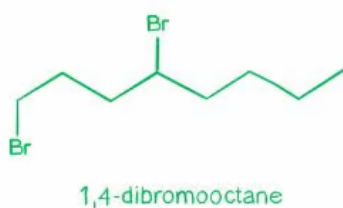
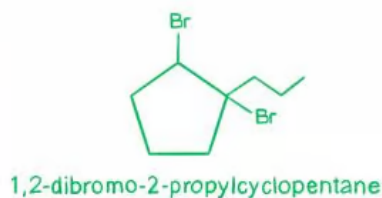
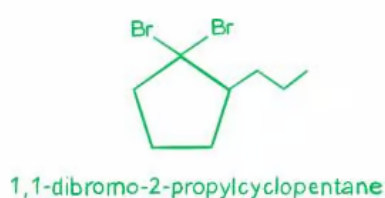


- Therefore, **C** is correct.

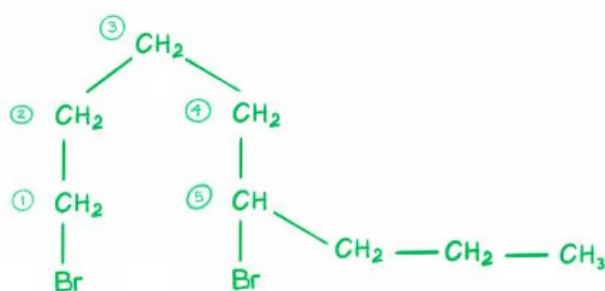
8

The correct answer is **D** because:

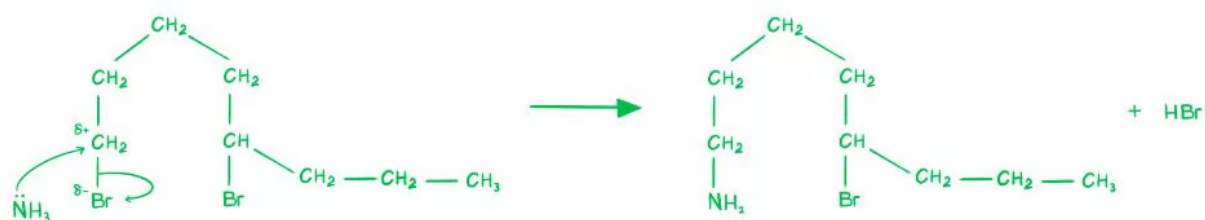
- First, let's draw out each molecule to get an idea of the structure of each.



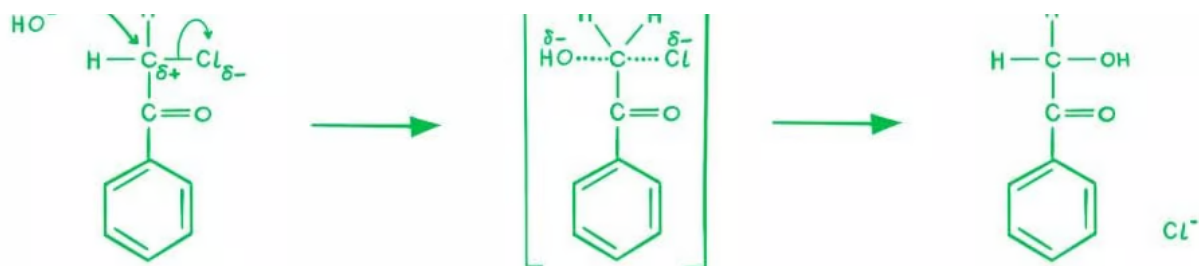
- Analysing the structure of each, we can see that 1,5-dibromooctane is the only one that could possibly make an aromatic ring.



- In this reaction, the bromines are substituted with ammonia via nucleophilic substitution.
- Ammonia will attack the carbon of one of the C-Br bonds with its lone pair, and so it'll replace Br with NH₂.



- And then, the NH₂ group can do the exact same thing with the other C-Br bond in order to form Coniine.

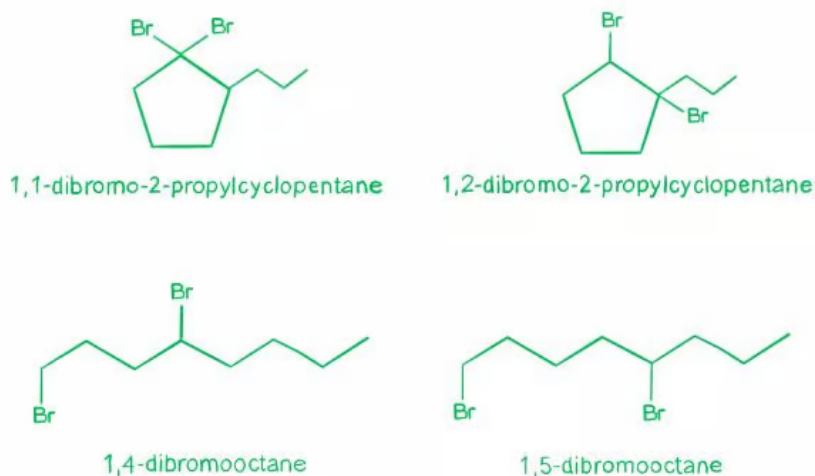


- Therefore, **C** is correct.

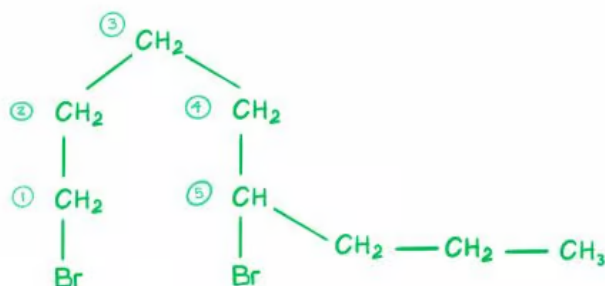
8

The correct answer is **D** because:

- First, let's draw out each molecule to get an idea of the structure of each.



- Analysing the structure of each, we can see that 1,5-dibromooctane is the only one that could possibly make an aromatic ring.



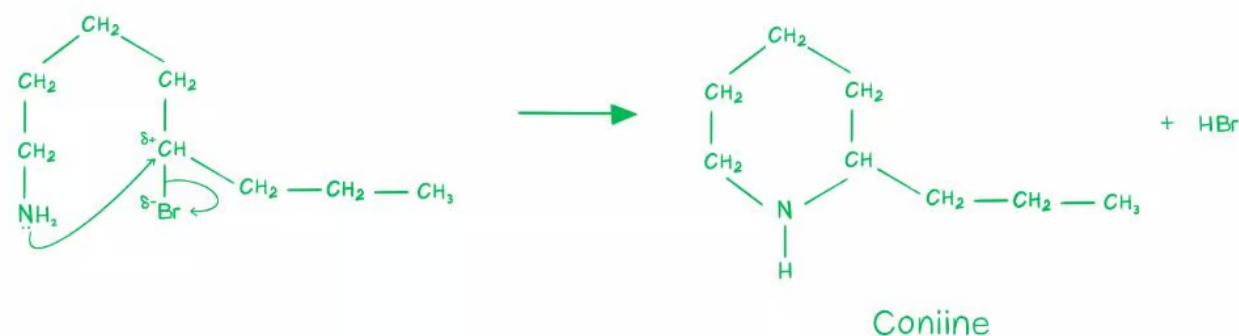
- In this reaction, the bromines are substituted with ammonia via nucleophilic

substitution.

- Ammonia will attack the carbon of one of the C-Br bonds with its lone pair, and so it'll replace Br with NH_2 .



- And then, the NH_2 group can do the exact same thing with the other C-Br bond in order to form Coniine.



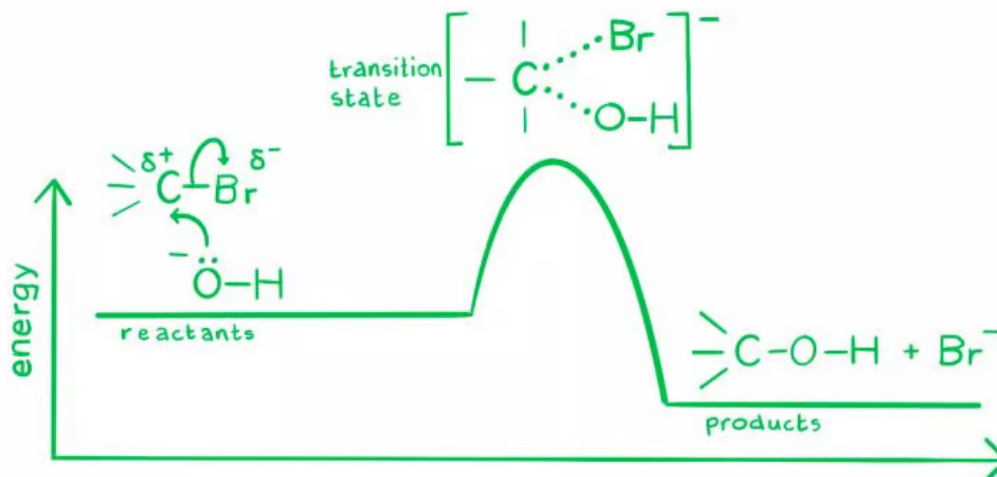
9

The correct answer is **B** because:

- Reaction pathway diagrams display information about the energy changes during chemical reactions.
- The reaction pathway diagram shows a tertiary halogenoalkane which undergoes the $\text{S}_{\text{N}}1$ mechanism for nucleophilic substitution reactions.
- The first bump corresponds to the slow step, as more energy is required here, and the second bump corresponds to the fast step, as less energy is required at this stage.
- The halogenoalkane $\text{C}_2\text{H}_5\text{Br}$ in option **B** is a primary halogenoalkane which

undergoes the S_N2 mechanism for nucleophilic substitution.

- This kind of reaction can be represented on a reaction pathway by a single bump, as shown below:



- The peak represents the formation of the transition state, where the outgoing Br and the incoming OH are half-bonded with the functional group carbon atom.
 - Therefore, option **B** is correct.

A, C & D are incorrect as these are all examples of S_N1 reactions, and follow the double bump reaction pathway.

10

The correct answer is **C** because:

- Compound **C** does not exhibit cis-trans isomerism due to the fact that the unsaturated carbon atoms are both bonded to 4 identical chlorine atoms.
- All differences in electronegativities cancel out, therefore, no overall permanent dipole can form.

A is incorrect as this compound is a halogenoalkane that can exhibit cis-trans isomerism. Due to this it may or may not have a permanent dipole.

B is incorrect as this compound has 1 carbon atom bonded to 3 chlorine atoms and 1 hydrogen atom. The difference in electronegativity between the hydrogen and carbon atom will not match the difference in electronegativity between the opposite chlorine

and carbon atom. Due to this mismatch the molecule will exhibit a partial charge and hence a permanent dipole.

D is incorrect as this compound has 5 hydrogen atoms bonded to a carbon atom and 1 chlorine atom bonded to a carbon atom, similar to the scenario in compound **B**, this will cause a permanent dipole.

